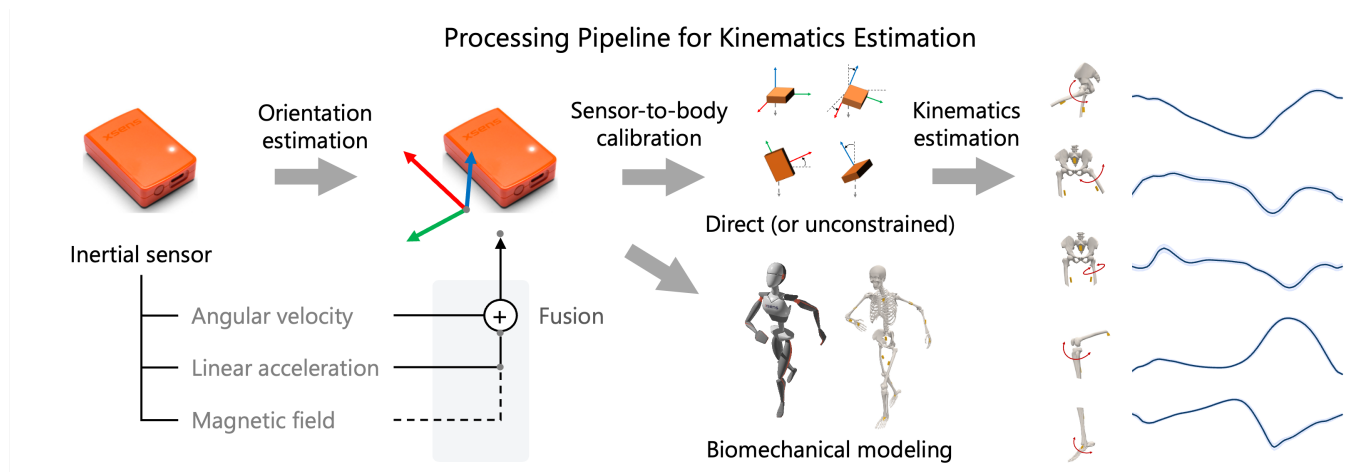




Inertial Motion Tracking Matches Marker-Based Tracking Accuracy: Rethinking Modeling Approaches Towards Future Progress

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IMULab is a workflow for state estimation and tracking of human movements with inertial measurement units (IMUs). A web app for users who may not wish to interact with the code is also being developed and will be released no later than the acceptance/publication of this manuscript.

Structure

See [imu_benchmark](#) for the implementation associated with Experiments 1, 2, and 3 presented in the paper; [imu_biodynamics](#) for Experiment 4.

Installation

The code is tested with **Python 3.10** on **Ubuntu 24.04** and **MacOS**. We recommend creating environments with **conda** to run the code.

```
conda create --name imu_benchmark python==3.10
conda activate imu_benchmark
```

See [AHRS](#), [VQF](#), and [RIANN](#) to install packages for state-estimation filters.

See [Scripting in Python](#) to install the **opensim** package for use of biomechanical modeling.

Run `python -m pip install -r requirements.txt` for other dependencies.

Data

The [complete dataset](#) is available for reviewers on FigShare, and will be published upon the acceptance/publication of the manuscript.

See [README.md](#) in the [imu_benchmark](#) or [imu_biodynamics](#) folder for how to store the data.

Run the Code

See [README.md](#) in the [imu_benchmark](#) or [imu_biodynamics](#) folder for how to run the code.